Practical scientific writing

Laboratory Work No. 3

NONOGA Djintoba

February 09, 2026

Objective of the Work

To become familiar with the LaTeX language and continue studying its capabilities.

Task

1. Launch several different programs, study a new package for working with graphics and new language commands.

Execution of Laboratory Work

Start working with a new package. To add graphics from an external source to LaTeX, we use the graphicx package, which adds the ' includegraphics command to LaTeX.

Figure 1: Program 1

Try changing the image's height and width. LaTeX automatically scales the image to preserve its aspect ratio.

Figure 2: Program 2

We can not only change the width and height of the image but also crop it and rotate it at various angles.

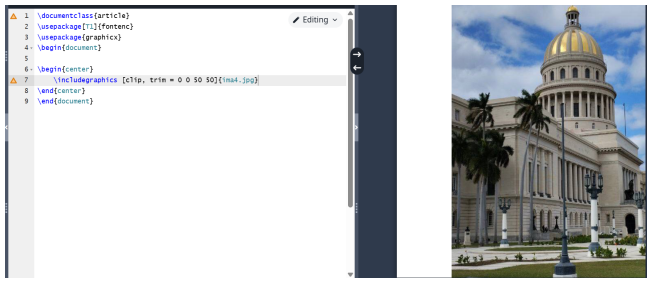

Figure 3: Program 3

Graphical elements can be moved to another place in the document. This is called a floating element. Let's try implementing this in a program.

Figure 4: Program 4

Often, it is required for a figure to appear in the output exactly where it is placed in the input data. The float package allows this to be done using the H option.

Figure 5: Program 5

If we need several environments, this can be done using the trivfloat package. It provides a single command trivfloat for creating new types of floating environments.

Figure 6: Program 6

For LaTeX to remember a place in the document, it needs to be marked and then referenced from other places.

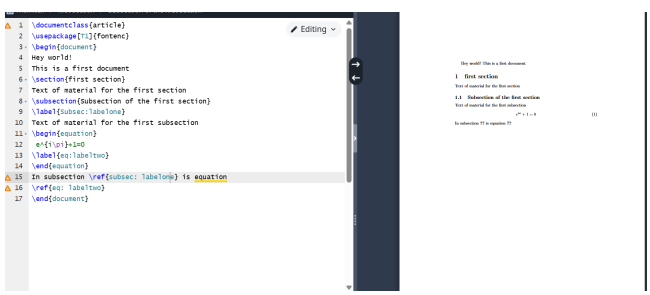

Figure 7: Program 7

Turn cross-references into hyperlinks using the hyperref package. It should be noted that if we place a reference outside of a chapter, it will neither display nor function.

Figure 8: Program 8

The programs work correctly.

Conclusions

I became familiar with the LaTeX language and continued studying its capabilities.